

Uniform m -Flat Pandrosion Atlases

Finite chart covers, anchor palettes, and uniform local contraction

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Abstract

The all-order tensor inverse-jet theorem gives a canonical m -flat chart near every regular zero of an analytic system. This paper globalizes that local statement over finite collections of regular zeros. An m -flat Pandrosion atlas is a finite family of local charts in which the transformed residuals satisfy

$$G_i(u) = u + \mathcal{O}(\|u\|^{m+1})$$

with uniform constants. In such an atlas, the anchored averaged-Jacobian Pandrosion map satisfies the uniform estimate

$$\|P_{G_i,a}(s)\| \leq C \|s\| (\|a\| + \|s\|)^m.$$

Consequently, if anchors and iterates remain in a chart ball of radius h , the raw charted Pandrosion map is a contraction with Lipschitz constant at most Ch^m . For analytic systems with finitely many regular zeros in a compact region, inverse jets produce a finite m -flat atlas after the neighborhoods are chosen small enough. The paper then formulates the dynamic atlas reconstruction conjecture: a chart-switching Pandrosion engine with equal-value pole avoidance and finite-slope conditioning control should asymptotically behave as if it were selecting from such a finite m -flat atlas.

Scope and status. This paper is the atlas continuation of the all-order tensor inverse-jet paper. The main theorems are local-uniform compactness statements. The final dynamic reconstruction principle is conjectural.

1 Why uniform atlases?

Local inverse jets solve one root at a time. A numerical method, however, does not begin with the root already identified. It moves through a phase space where several roots, charts, scales, and singular barriers coexist. The atlas viewpoint separates two tasks:

- (i) near a regular zero, use an m -flat chart to suppress the local multiplier;
- (ii) away from one chart, switch to another chart before conditioning or logarithmic scale becomes unstable.

The purpose of this paper is to state what a finite, uniformly controlled version of this picture gives.

2 m -flat atlas data

Let $G : U \subset \mathbb{C}^n \rightarrow \mathbb{C}^n$ be analytic. Assume G has finitely many regular zeros $\rho_1, \dots, \rho_N$ in a compact set $K \subset U$, meaning

$$G(\rho_j) = 0, \quad DG(\rho_j) \text{ is invertible.}$$

Definition 2.1 (*m*-flat Pandrosion chart). A chart $\phi_j : V_j \rightarrow U$ centered at ρ_j is *m*-flat if

$$\phi_j(0) = \rho_j, \quad G(\phi_j(u)) = u + \mathcal{O}(\|u\|^{m+1}).$$

It is *(m, C, r)*-controlled if $B(0, r) \subset V_j$ and, for all $\|u\| \leq r$,

$$\|G(\phi_j(u)) - u\| \leq C \|u\|^{m+1}, \quad \|D(G \circ \phi_j)(u) - I\| \leq C \|u\|^m.$$

Definition 2.2 (Uniform *m*-flat atlas). A finite family

$$\mathfrak{A}_m = \{(V_j, \phi_j)\}_{j=1}^N$$

is a uniform *m*-flat atlas for the regular zeros in K if each ϕ_j is *(m, C, r)*-controlled with the same constants C, r .

Theorem 2.3 (Finite inverse-jet atlas). Let G have finitely many regular zeros $\rho_1, \dots, \rho_N$ in K . For every $m \geq 1$, there exists a uniform *m*-flat atlas around these zeros.

Proof. At each regular zero, the all-order tensor inverse-jet theorem produces a polynomial chart $\phi_{j,m}$ satisfying

$$G(\phi_{j,m}(u)) = u + \mathcal{O}(\|u\|^{m+1}).$$

For each fixed j , analyticity gives constants $C_j, r_j > 0$ controlling the residual and derivative remainders on $B(0, r_j)$. Since the number of roots is finite, take

$$C = \max_j C_j, \quad r = \min_j r_j.$$

After shrinking r if necessary so that all chart maps are biholomorphic on their balls, the family is uniformly controlled. $\square$

3 Uniform anchored estimates

Inside chart j , write

$$F_j(u) = G(\phi_j(u)).$$

The averaged-Jacobian divided difference is

$$\mathcal{Q}_j(a, s) = \int_0^1 DF_j(s + t(a - s)) dt,$$

and the charted anchored Pandrosion map is

$$P_{j,a}(s) = s - \mathcal{Q}_j(a, s)^{-1} F_j(s).$$

Theorem 3.1 (Uniform two-point atlas estimate). Let $\mathfrak{A}_m$ be a uniform *(m, C, r)*-controlled atlas. There are constants $C_1 > 0$ and $0 < r_1 \leq r$, depending only on the atlas constants, such that for every chart j , every $\|a\|, \|s\| \leq r_1$, and every invertible $\mathcal{Q}_j(a, s)$,

$$\|P_{j,a}(s)\| \leq C_1 \|s\| (\|a\| + \|s\|)^m.$$

Proof. The proof is uniform version of the local two-point estimate. From the controlled atlas bounds,

$$F_j(s) = s + R_j(s), \quad \|R_j(s)\| \leq C \|s\|^{m+1},$$

and

$$\mathcal{Q}_j(a, s) = I + E_j(a, s), \quad \|E_j(a, s)\| \leq C' (\|a\| + \|s\|)^m$$

with C' independent of j . Choose r_1 so small that $\|E_j(a, s)\| \leq 1/2$. Then

$$\mathcal{Q}_j(a, s)^{-1} = I - E_j(a, s) + \mathcal{O}(\|E_j(a, s)\|^2)$$

with a uniform inverse bound. Substituting into $P_{j,a}(s) = s - \mathcal{Q}_j(a, s)^{-1} F_j(s)$ gives the estimate with a constant independent of j . $\square$

Corollary 3.2 (Uniform chart contraction). *Under the hypotheses of Theorem 3.1, suppose $\|a\| \leq h$, $\|s\| \leq h$, and $h \leq r_1$. Then*

$$\|P_{j,a}(s)\| \leq C_2 h^m \|s\|$$

uniformly in j . If $C_2 h^m < 1$, the charted anchored map is a contraction on the local chart ball.

Proof. Apply Theorem 3.1 and use $\|a\| + \|s\| \leq 2h$. $\square$

4 Anchor palettes

The monomial theory used scale palettes; the charted theory uses anchor palettes. In a local chart, an anchor palette at mesh h is a finite set $\mathcal{P}_h \subset B(0, h) \setminus \{0\}$ from which the algorithm chooses anchors.

Proposition 4.1 (Palette multiplier bound). *Let $\mathfrak{A}_m$ be a uniform m -flat atlas. If every selected anchor satisfies $0 < \|a\| \leq h$, then*

$$\|DP_{j,a}(0)\| \leq C_3 h^m$$

with C_3 independent of the chart and of the selected palette point.

Proof. The derivative version follows from the same uniform estimate applied to the derivative at $s = 0$, or directly from

$$DP_{j,a}(0) = I - \mathcal{Q}_j(a, 0)^{-1}.$$

Since $DF_j(u) = I + \mathcal{O}(\|u\|^m)$ uniformly in j , the result follows. $\square$

5 Singular set and chart switching

The atlas estimates are local. The obstruction to a global theorem is the singular set

$$\Sigma = \{\det DG = 0\} \cup \{\text{equal-value pole configurations}\} \cup \{\text{chart degeneracies}\}.$$

A dynamic engine should switch charts before the trajectory approaches Σ or before logarithmic scale becomes large.

Definition 5.1 (Atlas-admissible trajectory). *A trajectory is atlas-admissible if every iterate is represented in some chart of a finite atlas, the charted coordinate remains in a controlled ball, the selected anchor belongs to a bounded palette, and the equal-value separation and finite-slope condition number remain bounded away from their singular limits.*

Theorem 5.2 (Local uniform convergence under atlas admissibility). *Let an atlas-admissible trajectory eventually remain in a single m -flat chart with anchor radius h satisfying $C_2 h^m < 1$. Then the raw charted anchored Pandrosion iterates converge linearly to the chart root. The contraction factor is at most $C_2 h^m$.*

Proof. After the final chart switch, Corollary 3.2 applies at every step in the chart ball. The Banach fixed-point argument gives convergence to the unique fixed point in that ball, namely the chart coordinate 0 of the regular zero. $\square$

6 Dynamic reconstruction conjecture

Conjecture 6.1 (Uniform m -flat atlas reconstruction). *Let G be analytic with finitely many regular zeros in a compact nonsingular region. A Pandrosion engine that combines dynamic chart switching, equal-value pole avoidance, finite-slope conditioning filters, and symmetric finite-probe inverse-jet estimation asymptotically selects from a finite uniform m -flat atlas along every regular convergent trajectory.*

Conjecture 6.2 (Atlas contraction outside transition times). *For a regular trajectory that avoids Σ , there is a finite set of chart-transition times outside of which the normalized Pandrosion map is locally contractive in the active m -flat chart. The effective contraction factor is controlled by the active anchor radius as $\mathcal{O}(h^m)$.*

7 Conclusion

The all-order inverse-jet theorem is local; the uniform m -flat atlas theorem is finite and geometric. It says that once the regular roots are covered by controlled inverse-jet charts, the raw anchored Pandrosion map has a uniform local contraction factor of order h^m . This gives a clean mathematical target for dynamic chart engines: construct, approximate, and switch among such charts without direct access to the exact roots.

References

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